



Capacitive_touch_and_led

PROJECT DOCUMENT

Generated by: Bruno Santos, brunosantos67040@gmail.com

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PROJECT DETAILS

Project name:

Capacitive_touch_and_led

Project description:

Author: Bruno Santos Project: Capacitive Touch + LED for ESP32 DOIT DevKit V1
Touch Pin: GPIO 4 (T0) Built-in LED: GPIO 2

Selected CAD tool:

Altium Designer

Generated by:

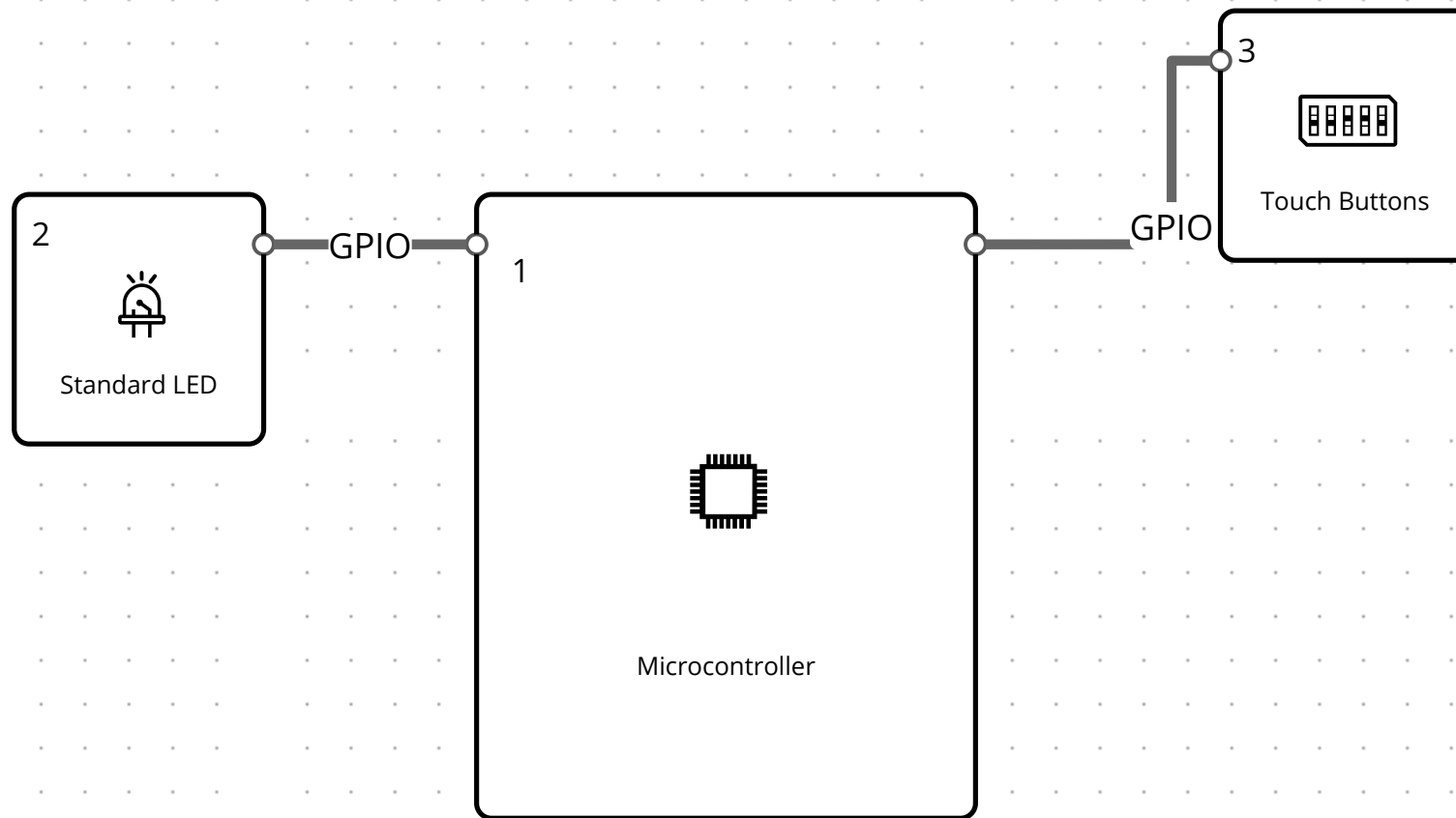
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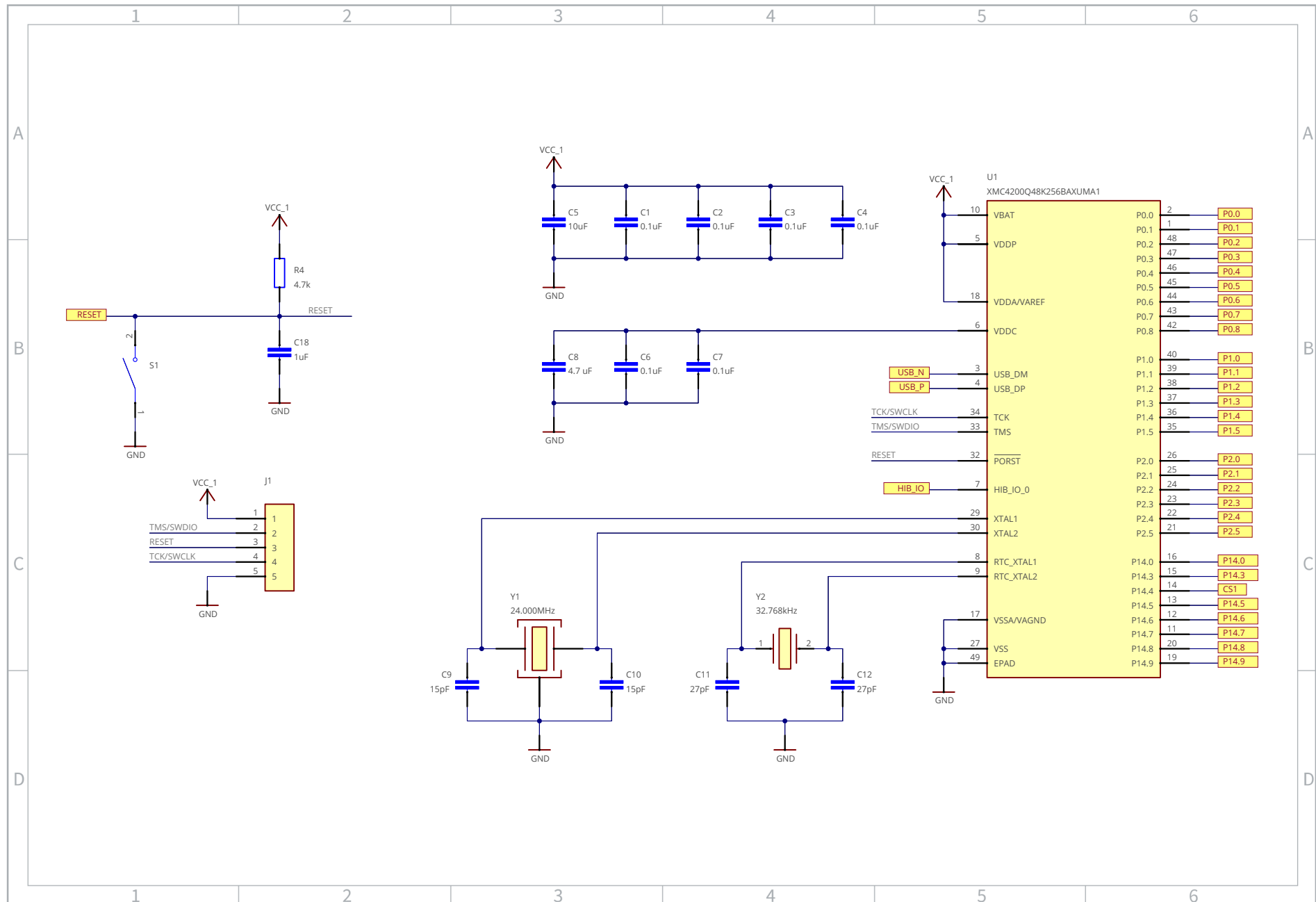
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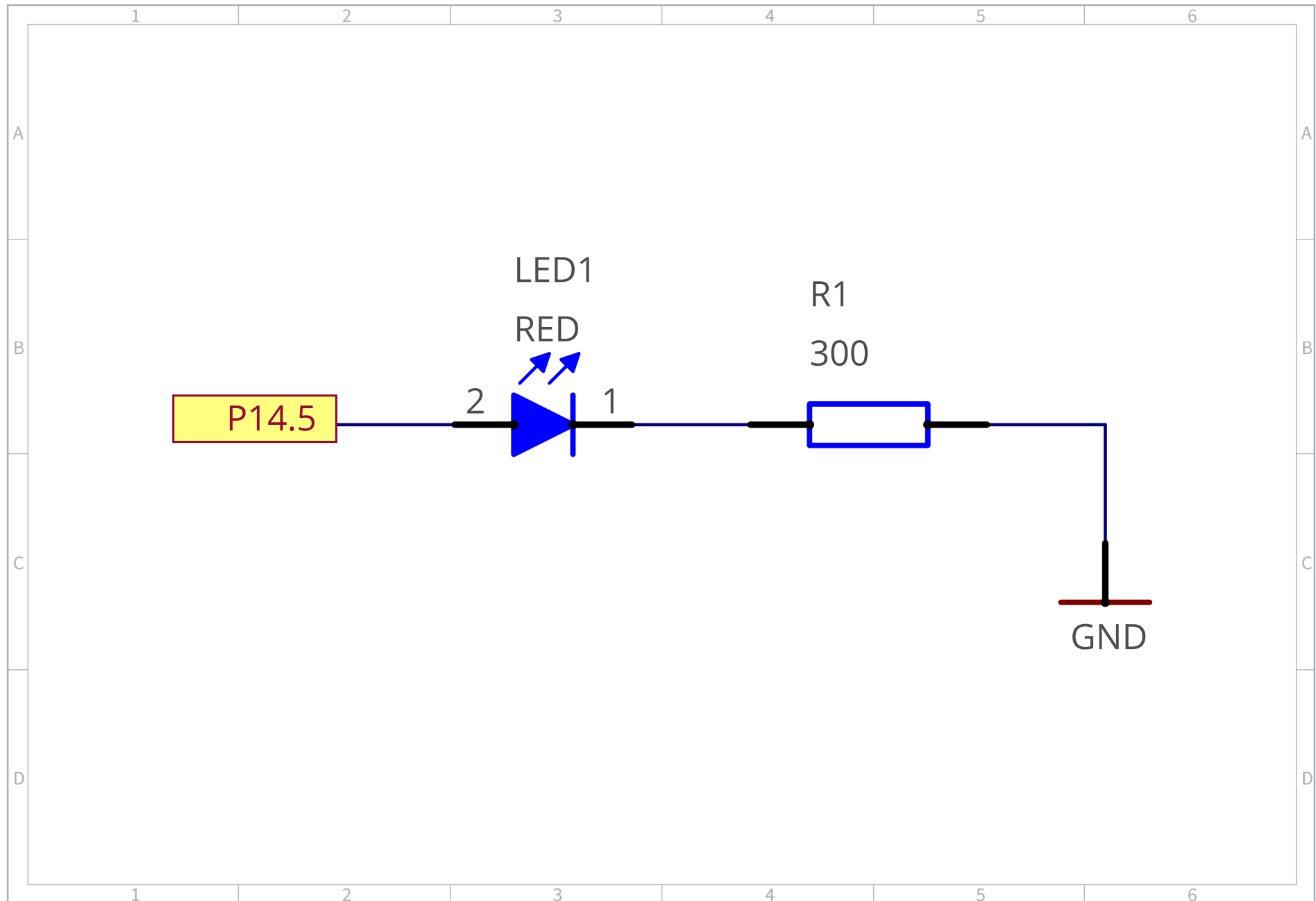
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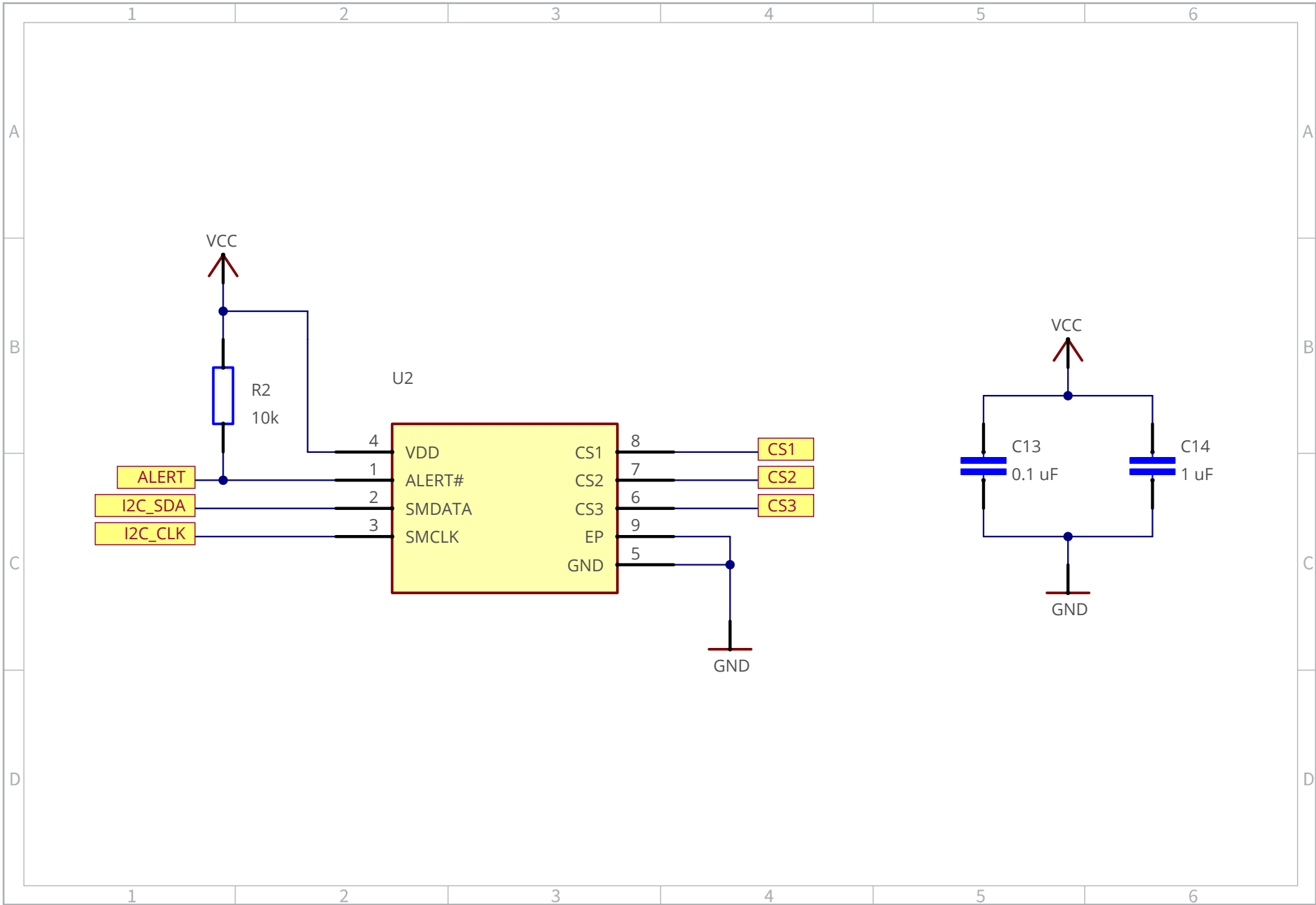
DISCLAIMER

Project Title, Project Description, and Block Diagram may be AI-Generated.
Schematics, BOM, and Components are not AI-Generated.









Ref. Des.	Value	Manufacturer	MPN	Package	Description
C1	100 nF	KYOCERA AVX COMPONENTS CORP	06035C104K4T2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603
C6	100 nF	KYOCERA AVX COMPONENTS CORP	06035C104K4T2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603
C3	100 nF	KYOCERA AVX COMPONENTS CORP	06035C104K4T2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603
C2	100 nF	KYOCERA AVX COMPONENTS CORP	06035C104K4T2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603
C4	100 nF	KYOCERA AVX COMPONENTS CORP	06035C104K4T2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603
C7	100 nF	KYOCERA AVX COMPONENTS CORP	06035C104K4T2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0603
C5	10 µF	KYOCERA AVX COMPONENTS CORP	06036D106MAT2A	0603	Ceramic Capacitor, Multilayer, Ceramic, 6.3V, 20% +Tol, 20% -Tol, X5R, 15% TC, 10uF, Surface Mount, 0603
C8	4.7 µF	SAMSUNG ELECTRO-MECHANICS	CL05A475KP5ZRNC	0402	Ceramic Capacitor, Multilayer, Ceramic, 10V, 10% +Tol, 10% -Tol, X5R, 15% TC, 4.7uF, Surface Mount, 0402
C10	15 pF	YAGEO CORP	CC0402JRNPO9BN150	0402	Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000015uF, Surface Mount, 0402
C9	15 pF	YAGEO CORP	CC0402JRNPO9BN150	0402	Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000015uF, Surface Mount, 0402
C12	27 pF	YAGEO CORP	CC0402JRNPO9BN270	0402	Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000027uF, Surface Mount, 0402
C11	27 pF	YAGEO CORP	CC0402JRNPO9BN270	0402	Ceramic Capacitor, Multilayer, Ceramic, 50V, 5% +Tol, 5% -Tol, C0G, 30ppm/Cel TC, 0.000027uF, Surface Mount, 0402
C18	1 µF	KYOCERA AVX COMPONENTS CORP	KGM15BR51C105KT	0603	Ceramic Capacitor, Multilayer, Ceramic, 16V, 10% +Tol, 10% -Tol, X5R, 15% TC, 1uF, Surface Mount, 0603
J1	MALE	WUERTH ELEKTRONIK GMBH & CO KG	61300511021		Board Connector, 5 Contact(s), 1 Row(s), Male, Right Angle, 0.1 inch Pitch, Solder Terminal, Locking, Black Insulator, Plug
R4	4.7 kOhm	PANASONIC CORP	ERA2AED472X	0402	Fixed Resistor, Thin Film, 0.063W, 4700ohm, 50V, 0.5% +/-Tol, 25ppm/Cel, Surface Mount, 0402

Ref. Des.	Value	Manufacturer	MPN	Package	Description
S1		OMRON CORP	B3U-1000P		Keypad Switch, 1 Switches, SPST, Momentary, 0.05A, 12VDC, 1.99N, Solder Terminal, Surface Mount-straight
U1		INFINEON TECHNOLOGIES AG	XMC4200Q48K256BAXUMA1		RISC Microcontroller, 32-Bit, FLASH, 80MHz, CMOS, PQCC48
Y1	24 MHz	DIODES INC	FL2400022		Parallel - Fundamental Quartz Crystal, 24MHz Nom
Y2	32.768 kHz	ABRACON LLC	ABS06-32.768KHZ-1-T		Parallel - Fundamental Quartz Crystal, 0.032768MHz Nom
LED1		WUERTH ELEKTRONIK GMBH & CO KG	151053RS03000		Single Color LED, Red, Water Clear, T-1 3/4, 5mm
R1	300 Ohm	YAGEO CORP	RC0603FR-07300RL	0603	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 300ohm, 75V, 1% +/-Tol, 100ppm/Cel, Surface Mount, 0603
C13	100 nF	SAMSUNG ELECTRO-MECHANICS	CL05B104KA5NNNC	0402	Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X7R, 15% TC, 0.1uF, Surface Mount, 0402
C14	1 µF	SAMSUNG ELECTRO-MECHANICS	CL05X105KA5NQNC	0402	Ceramic Capacitor, Multilayer, Ceramic, 25V, 10% +Tol, 10% -Tol, X6S, 22% TC, 1uF, Surface Mount, 0402
R2	10 kOhm	PANASONIC CORP	ERJ2GEJ103X	0402	Fixed Resistor, Metal Glaze/thick Film, 0.1W, 10000ohm, 50V, 5% +/-Tol, 200ppm/Cel, Surface Mount, 0402
U2		MICROCHIP TECHNOLOGY INC	CAP1203-1-AC3-TR		Analog Circuit, 1 Func, 3 Channel, PDSO8